Clustering Approaches to Adjust for Population Stratification

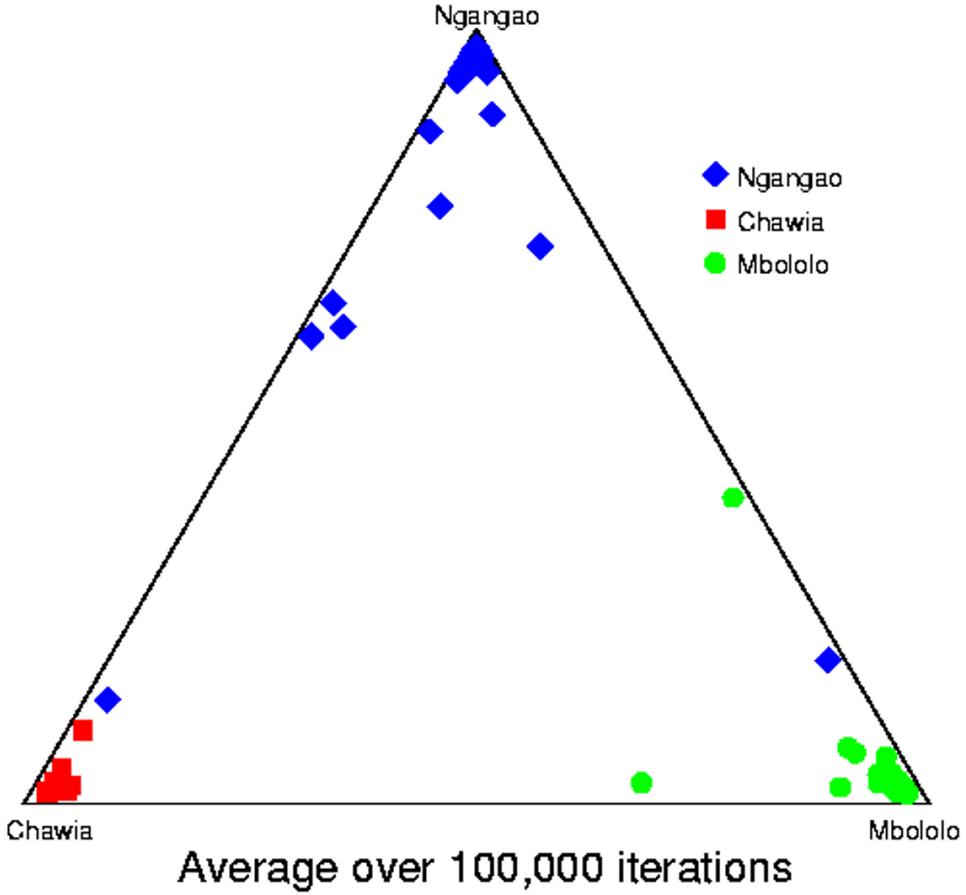
STRUCTURE

- Bayesian clustering model (Pritchard, Stephens, Donnelly, Genetics 2000) **infer the latent population structure** by a stratified analysis.
- Assigns individuals to K source populations; under admixture, each individual can have **fractional ancestry** across populations.
- Works best with **strong population structure** (a few distinct ancestries) and **many variants / ancestry-informative markers (AIMs)**.
- A practical issue is choosing K in advance.
  - Larger K can overfit, so model-selection/diagnostic procedures are needed.
  - Often used exploratorily by comparing results across several values of K.

STRUCTURE

Convergence of Structure: Taita thrush data (K=3).

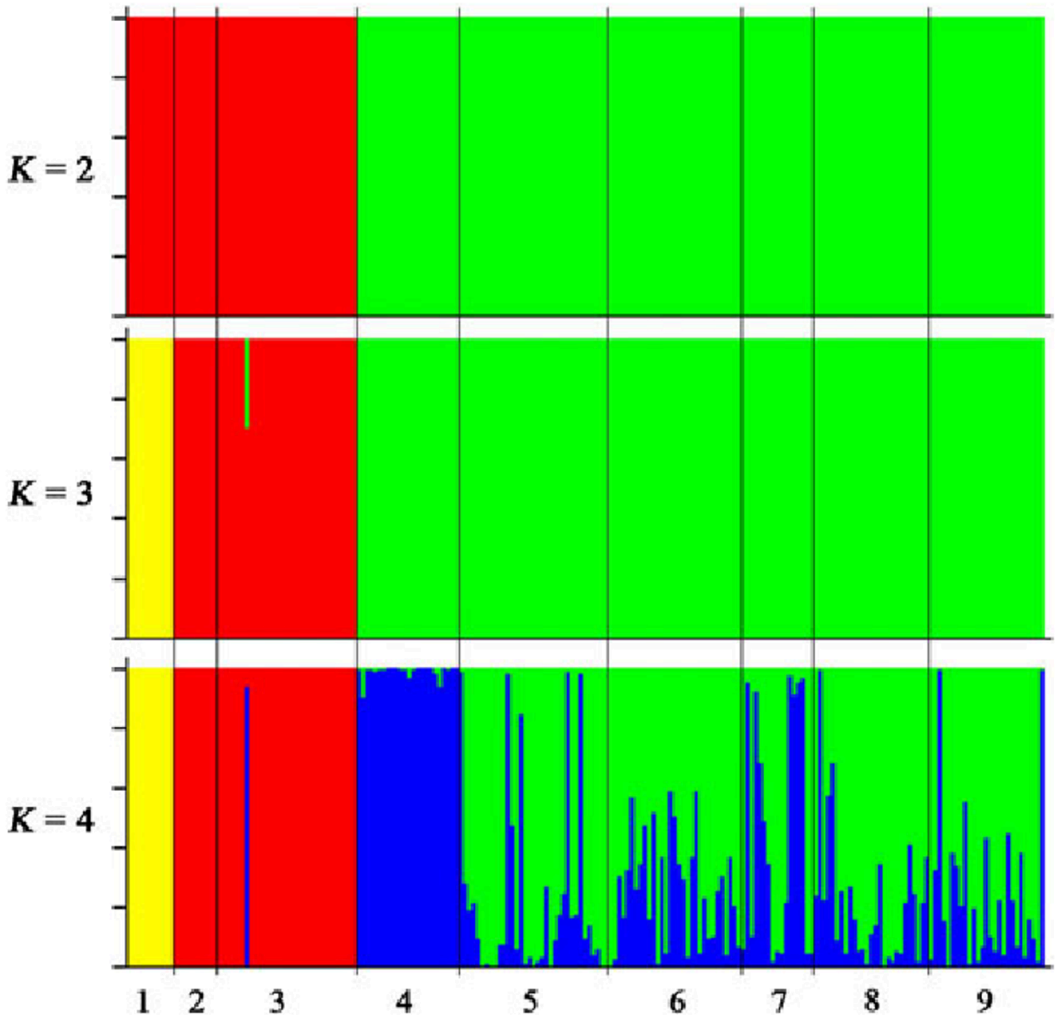
- The ternary plot shows each individual's proportional ancestry.
- Points near a vertex indicate near-pure membership in one cluster; interior points indicate admixture.



Source: Publicly available from Google Images

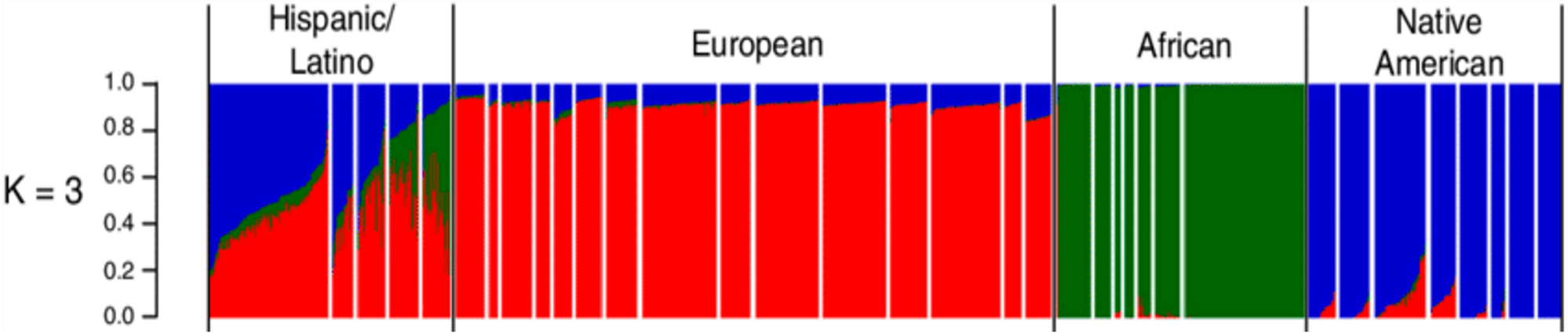
STRUCTURE Bar Plots

- Each vertical bar = one individual; the color fractions are the estimated membership coefficients.
- Individuals with same color patterns are inferred to belong to the same cluster.



Source: Publicly available from Google Images

Applying STRUCTURE to Admixed Populations



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